METRIC

MILITARY SPECIFICATION

PAINT, ALUMINUM, HEAT-RESISTING (650°C), LOW-EMISSIVITY (0.40 OR LESS) (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two compositions of aluminum pigmented, heat-resistant, low-emissivity paint that will withstand solvents, normal weather exposure, and temperatures up to 650° Celsius (C), and will maintain an emissivity factor below 0.40.

1.2 Classification. Heat-resisting aluminum paint covered by this specification shall be of the following compositions, as specified (see 6.2):

Composition G - General use.
Composition L - Limited use (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-P-320 - Pigment, Aluminum; Powder and Paste for Paint.
TT-S-735 - Standard Test Fluids; Hydrocarbon.
TT-T-306 - Thinner, Synthetic Resin, Enamel.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

STANDARDS

FEDERAL
FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED-STD-313 - Symbols for Packages and Containers for Hazardous Industrial Chemicals and Materials.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 569-72 - Steel, Carbon (0.15 maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality.
D 93-73 - Flash Point by Pensky-Martens Closed Tester.
D 562-55 - Consistency of Paints Using the Stormer Viscosimeter.
D 610-68 - Rusting on Painted Steel Surfaces, Evaluating Degree of.
D 1014-66 - Exterior Exposure Tests of Paints on Steel, Conducting.
D 1308-57 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes.
D 2369-73 - Volatile Content of Paints.
D 2698-73 - Pigment Content of Solvent-Type Paints by High Speed Centrifuging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The paint furnished under this specification shall consist of ingredients conforming to the quantitative requirements specified in table I when tested in accordance with 4.3.

TABLE I. Quantitative requirements.

Characteristics	Requirements Minimum	Maximum
Total solids, percent by weight of paint	40	--
Pigment, percent by weight of paint	23	27
Vehicle solids, percent by weight of paint	13	--
Silica, (SiO_2), percent by weight of vehicle solids	11	--
Consistency (package) Krebs Stormer shearing rate 200 RPM, grams	75	125
Equivalent Krebs Units (KU)	54	67
Drying time, hours:		
Set to touch, air dry	--	2
Dry hard, air dry	--	6
Full hardness, baked at 204°C	--	1
Emissivity	--	0.40
Flash point, °C	38	--

3.2 Composition.

3.2.1 Pigment. The pigment shall be aluminum conforming to type I or type II, class A of TT-P-320.

3.2.2 Vehicle. The vehicle shall be a silicone or modified silicone resin together with driers, if necessary, and volatile aromatic or aliphatic-type solvents to meet the requirements of this specification.

3.2.2.1 Solvent (composition L only). The solvent portion of composition L shall conform to the following requirements when tested as specified in 4.3.4 (see 6.4).

(a) Aromatic compounds with eight or more carbon atoms except ethyl benzene: 8 percent maximum.
(b) Ethyl benzene and toluene: 20 percent maximum.
(c) Solvents with an olefinic or cyclo-olefinic type of unsaturation: negative test.
(d) Ketones: negative.
(e) Total of (a) + (b): 20 percent maximum.

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3.3 Qualitative requirements. The paint shall meet the following qualitative requirements:

3.3.1 Condition in container. A freshly opened full container of the paint tested as specified in 4.3.6 shall be free from grit, seeds, skins, lumps, abnormal thickening, or livering, and shall show no more pigment settling or caking than can be readily reincorporated by hand to a smooth homogeneous state.

3.3.2 Storage stability.

3.3.2.1 Partially full container. A three-quarter filled, closed 8-ounce glass jar of the paint shall show no skinning when tested as specified in 4.3.7.1. After aging as specified in 4.3.7.1, the paint shall show no livering, curdling, hard caking, or tough gummy sediment. It shall mix readily to a smooth uniform state, and any skin formed shall be continuous and easily removed.

3.3.2.2 Full container. A full quart can of the paint shall show no gas pressure in the can, no skinning, livering, curdling, hard dry caking, or tough gummy sediment when tested as specified in 4.3.7.2. The paint shall mix readily to a smooth uniform state. There shall be no evidence of fungus or mildew, their growth having been prevented by an organic, non-mercurial fungicide.

3.3.3 Brushing properties. The brushing properties when tested as specified in 4.3.8 shall produce a smooth uniform film free from seeds, runs, sags, or streaks.

3.3.4 Spraying properties. The paint, tested as specified in 4.3.9, shall spray in all respects as required herein and shall show no running, sagging, or streaking. The dried film shall show no dusting or mottling and shall present a smooth uniform finish free from seeds.

3.3.5 Heat resistance. Films of paint prepared and tested as specified in 4.3.10 shall show no cracking, blistering, flaking, or peeling. When cut as specified in 4.3.10 the paint shall adhere tightly to the metal.

3.3.6 Salt spray resistance. Films of paint prepared and tested as specified in 4.3.11 and examined immediately after removal from the test shall show no more than a trace of rusting and no more than 5 scattered blisters none of which is larger than 1 millimeter (mm) in diameter. Dulling or staining unaccompanied by red rusting shall be permitted. On removal of the paint from the panels tested as specified in 4.3.11.2, the surface of the steel shall show deterioration no greater than grade 8 of ASTM D 610-68.

3.3.7 Water resistance. A film of paint prepared and tested as specified in 4.3.12 shall show no wrinkling or blistering immediately upon removal of the panel from the water. The paint shall be no more than slightly affected when examined 2 hours after removal; and after 24 hours air drying, the portion of the paint which was immersed shall show no more than a slight whitening or dulling in comparison to the portion that was not immersed.

3.3.8 Hydrocarbon fluid resistance. A film of the paint prepared and tested as specified in 4.3.13 shall show no wrinkling or blistering immediately after removal from the fluid. After 24 hours air drying, the portion of the panel which was immersed shall show no discernible difference with regard to hardness and adhesion from a panel prepared at the same time but not immersed.

3.3.9 Weather resistance A film of paint prepared and tested as specified in 4.3.14 shall show no checking, cracking, or film deterioration. Dulling or staining unaccompanied by red rusting shall be permitted. On removal of the paint from the panels the surface of the steel shall show deterioration no greater than grade 8 of ASTM D 610-68.

3.3.10 Toxicity. The paint shall contain no benzene (benzol), methanol, chlorinated compounds, hydrolyzable chlorine derivatives, or other ingredients which are deemed toxicologically hazardous under normal conditions of usage.

3.3.11 Emissivity. Test panels prepared, painted, and tested as specified in 4.3.10.1 shall exhibit emissivity characteristics not exceeding 0.40 when determined in accordance with 4.3.16.

3.3.12 Flash point. The material shall have a minimum flash point of 38°C when tested in accordance with 4.3.17.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance testing of individual lots shall consist of all the tests of 4.3 with the exception of storage stability (see 4.3.7) and weather resistance (see 4.3.14).

4.2.1 Sampling and inspection. Sampling and inspection shall be in accordance with method 1031 of FED-STD-141.

4.3 Test procedures.

4.3.1 Test conditions. The routine and referee testing conditions shall be in accordance with section 7 of FED-STD-141 except as otherwise specified herein. The dry film thickness shall be measured after the coated panel has been baked at 204°C plus 2°C for 15 minutes.

4.3.2 The tests listed in table II shall be conducted in accordance with FED-STD-141 or as required by this specification.

TABLE II. Tests.

Item	Test method			Requirements
	Applicable Method of FED-STD-141	ASTM No.	Test Paragraph	
Total solids		D 2369-73	--	Table I
Pigment content		D 2698-73	--	Table I
Isolation of vehicle		D 2698-73	--	--
Vehicle solids		D 2698-73	--	Table I
Silica	--		4.3.3	Table I
Consistency		D 562-55	--	Table I
Drying time:				
Set to touch	4061		4.3.5.1	Table I
Dry hard	4061		4.3.5.1	Table I
Full hardness	--		4.3.5.2	Table I
Condition in container	3011		4.3.6	3.3.1
Storage stability:				
Partially full container	3021		4.3.7.1	3.3.2.1
Full container	3022		4.3.7.2	3.3.2.2
Brushing properties	4321		4.3.8	3.3.3
Spraying properties	4331		4.3.9	3.3.4
Heat resistance:				
Hot rolled steel	--		4.3.10.1	3.3.5
Cold rolled steel	--		4.3.10.2	3.3.5
Salt spray resistance:				
Hot rolled steel	6061		4.3.11.1	3.3.6
Cold rolled steel	6061		4.3.11.2	3.3.6
Water resistance		D 1308-57 Sec. 5D	4.3.12	3.3.7
Hydrocarbon fluid resistance		D 1308-57 Sec. 5D	4.3.13	3.3.8
Weather resistance		D 1014-66	4.3.14	3.3.9
Benzene	5091		4.3.15	3.3.10
Methanol	5133		4.3.15	3.3.10
Chlorinated solvents	5132		4.3.15	3.3.10
Emissivity	--		4.3.16	3.3.11
Flash point (°C)		D 93-73	4.3.17	3.3.12

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4.3.3 Silica content of vehicle. From a stoppered bottle or weighing pipet, weigh accurately by difference, about 3 grams of vehicle into a previously ignited and weighed 7.6 centimeters (cm) porcelain evaporating dish. Dry at 105°C, in an oven for 3 hours. Place the dried sample in a cold muffle furnace and gradually increase the temperature, over a period of 3 hours to 900°C, then maintain this temperature for an additional hour. After cooling in a dessicator, weigh the dish and contents and calculate the percent of silica as follows:

$$\text{Percent silica} = \frac{\text{Weight of ash} \times 100}{\text{Weight of sample} \times \text{non-volatile vehicle fraction}}$$

4.3.4 Solvent analysis for composition L.

4.3.4.1 Separation of volatile portion. Pour about 15 grams of the enamel into a 50 milliliter (ml) distilling flask. Add 10 ml of tricresyl phosphate and several anti-bumping stones or Berl saddles. Fit a release valve into the mouth of the flask and attach a delivery tube to the side arm, extending into a receiver. The receiver consists of a test tube (20 by 150 mm) with side arm for attaching to a vacuum pump. The glass delivery tube should reach 38 mm from the bottom of the tube. Immerse the receiver in a dry ice-acetone bath. Preheat a silicone oil bath to 180°C. Raise the oil bath until the oil reaches the sample level. Reduce the pressure slowly to 10 mm of mercury. After all solvent has distilled, carefully release the vacuum using the valve that is connected to the distilling flask. Reserve the collected distillate for the aromatic solvent determination and the test for ketone, olefinic and cyclo-olefinic compounds.

4.3.4.2 Determination of aromatic hydrocarbons.

Apparatus: A gas chromatograph equipped with a thermal conductivity detector.

Column preparation: Two lengths of 6.4 mm copper tubing, 1.83 meter (m) and 5.51 m long, are packed with 35 percent N, N-Bis(2-cyanoethyl) formamide on 60- to 80- mesh Chromosorb P.

Operating conditions:	1.83 m	5.51 m
Detector cell temperature, °C	300	300
Detector cell current, ma	150	150
Injection port temperature, °C	300	300
Helium flow at exit, cc/minute	175	110
Column temperature, °C	125	70

4.3.4.2.1 Aromatic and oxygenated solvents - procedure A. Install the 1.83 m column and follow the operating conditions described above. Inject about 3×10^{-6} liter of the isolated distillate and scan the chromatogram. The aliphatic solvents will emerge within 1 minute and the complete chromatogram should develop in about 5 minutes. From the position of the peaks observed on the chromatogram, select an internal standard that will be free of interference, such as cyclopentanol or cyclohexanol. Add 0.6 ml of internal standard to 3 ml of the distillate, analyze according to the above procedure. Peaks emerging after 1 minute are aromatic solvents along with any oxygenated solvents that may be present. Calculate the percent of aromatic and oxygenated solvents as follows:

$$\text{Percent aromatic and oxygenated solvents, v/v} = \frac{20^{1/} \times A}{1.02^{2/} \times B}$$

where, A = area of aromatic and oxygenated solvents
B = area of internal standard
1/ is percent of internal standard added
2/ is correction factor if cyclopentanol is used. If another internal standard is used, calibrate to determine the correction factor

NOTE: If the above determination exceeds 8 percent, continue with the following procedure:

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4.3.4.2.2 Total aromatic content - procedure B. Place 5 ml of the distillate in a 10 ml glass stoppered graduate. Add 5 ml of 85 percent sulfuric acid slowly while the graduate is being cooled with tap water. After the acid has been added, shake vigorously for 2 minutes then allow the layers to separate. Remove as much of the top layer as possible and wash with distilled water. Carefully pipet 3 ml of the washed solvent into a small flask followed by 0.6 ml of the internal standard. Mix and analyze according to procecure A. Calculate the percent of aromatics after acid treatment in the same manner as in procedure A and the percent of total aromatic solvents as follows:

$$\text{Percent total aromatic solvents, v/v} = \frac{V \times (100-A)}{100-B}$$

where, A = percent of aromatic and oxygenated solvents from procedure A
B = percent of aromatic solvents after acid treatment.

NOTE: If the total aromatic content of the solvent is between 8 percent and 20 percent, continue with the following procedure:

4.3.4.2.3 Toluene and ethylbenzene - procedure C. Install the 5.51 m column and follow the operating conditions described for that column. Add 0.3 ml of high purity benzene to the 3 ml sample used in procedure A. If the results of procedures A and B indicated the presence of oxygenated solvents, treat this sample with 85 percent sulfuric acid (use 3 ml acid) as described in procedure B. Inject about $3X10^{-6}$ liter of sample and allow the chromatograph to develop until all of the xylene isomers appear. Purge that column by raising the column temperature to 120°C. After the high boiling materials emerge, reset the column temperature to 70°C. Calculate the percent of toluene and ethylbenzene as follows:

$$\text{Percent toluene, v/v} = \frac{(\text{area of toluene peak})\ (1.017^{1/})\ (10)^{2/}}{(\text{area of benzene peak})}$$

$$\text{Percent ethylbenzene, v/v} = \frac{(\text{area of ethylbenzene peak})\ (1.054)^{1/}\ (10)^{2/}}{(\text{area of benzene peak})}$$

1/ is the correction factor for the detector response
2/ is the percentage of internal standard added

NOTE: Sensitivity of the instrument should be adjusted to keep peaks from running off the scale. Appropriate corrections must be made for changes in sensitivity when computing the peak areas.

4.3.4.3 Test for olefinic or cyclo-olefinic compounds. Take two test tubes and place two drops of the distillate in each. Dissolve the first sample in 1 ml of carbon tetrachloride and add one drop of one percent bromine in carbon tetrachloride. Shake and allow to set for 5 minutes. A positive test is indicated by the complete absence of yellow color when observed against a white background. Dissolve the second sample in 1 ml of acetone and add one drop of one percent permanganate solution (1 gram of potassium permanganate crystals in 95 ml of acetone and 5 ml of water). Shake and allow to set for 2 minutes. A positive test is indicated by the decolorization of the purple solution. The solvent is considered to fail the test for olefinic and cyclo-olefinic compounds if either of the above tests is positive (see 3.2.2 and 6.5).

4.3.4.4 Test for ketones.

4.3.4.4.1 Reagent. Two grams of 2,4-dinitrophenylhydrazine plus 4 ml of concentrated sulfuric acid plus 30 ml methanol (add slowly) plus 10 ml water.

4.5.4.4.2 Procedure. Pipet 1 ml of reagent into a 20 by 170 mm test tube. Add 10 drops of distillate and shake for 30 seconds. A yellow precipitate or cloud in the reagent layer indicates the presence of ketones. Run a blank using 1 ml of reagent and 10 drops of mineral spirits.

4.3.5 Drying time.

4.3.5.1 Air drying. Draw down a film of the paint on a glass panel using a 0.038 mm applicator (0.076 mm gap clearance) and determine air drying time under referee conditions in accordance with method 4061 of FED-STD-141 for compliance with table I.

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4.3.5.2 Full hardness. Determine full hardness by spraying a film of paint to a dry film thickness of $20.3X10^{-3}$ mm to $25.4X10^{-3}$ mm on a steel panel that has been solvent cleaned with the aliphatic naphtha-ethylene glycol monoethyl ether mixture in accordance with method 2011 of FED-STD-141. Air dry 30 minutes and then bake at 204°C plus 2°C for 1 hour. Determine compliance with full hardness requirement of table I. The film shall be considered to have reached full hardness when it is impossible to remove with a knife blade.

4.3.6 Condition in container. Determine package condition on acceptance testing in accordance with method 3011 of FED-STD-141 and observe for compliance with 3.3.1. Evaluate pigment settling or caking by proceeding as in method 3011 of FED-STD-141 but do not stir. Reseal and agitate the can for 3 minutes on a paint shaker[1/]. On re-examination of the contents, the disclosure of any gel bodies or undispersed pigment indicates unsatisfactory settling properties.

4.3.7 Storage stability.

4.3.7.1 Partially full container. Determine skinning after 48 hours in accordance with method 3021 of FED-STD-141 and observe for compliance with 3.3.2.1. Reseal and age for seven days at 60°C and observe for compliance with 3.3.2.1.

4.3.7.2 Full container. In accordance with method 3022 of FED-STD-141, allow a full standard quart can of the paint to stand undisturbed for one year and then examine the contents. Evaluate pigment settling or caking as in 3.3.1 except agitate the can for 5 minutes on the paint shaker prior to re-examination. Check for compliance with 3.3.2.2.

4.3.8 Brushing properties. Apply the paint as packaged using a 2-1/2 inch brush and observe for brushing properties in accordance with method 4321 of FED-STD-141 for compliance with 3.3.3. As a referee test for sagging use method 4494 of FED-STD-141 except make the drawdown a minimum of 254 mm long on clear plate glass. The 0.102 mm strip shall not make contact with the next thicker strip at any point within the 140 mm central portion of the blade path.

4.3.9 Spraying properties. Spray the paint as packaged or thinned with not more than five percent by volume of thinner conforming to TT-T-306 on a steel panel to a dry film thickness between $20.3X10^{-3}$ mm and $25.4X10^{-3}$ mm and observe for spraying properties in accordance with method 4331 of FED-STD-141 for compliance with 3.3.4. For referee test use automatic application in accordance with method 2131 of FED-STD-141.

4.3.10 Heat resistance.

4.3.10.1 Hot rolled steel. Select ten panels which are completely free of loose mill scale and have been cut to 76 mm by 152 mm from 14 gauge as rolled commercial quality low carbon steel conforming to ASTM A 569-72. Solvent clean with the aliphatic naphtha-ethylene glycol monoethyl ether solvent mixture of method 2011 of FED-STD-141. Spray the paint (one or two coats as necessary) on all eight panels to a total dry film thickness between $20.3X10^{-3}$ mm and $25.4X10^{-3}$ mm and air dry for at least 24 hours. Place the panels in an oven on a rack or holder so that no part of the panels is in direct contact with the bottom or sides of the oven and then subject to the following heating schedule:

	°Celsius
8 hours (first day)	200
16 hours (overnight)	260
8 hours	320
16 hours	380
8 hours	440
16 hours	500
8 hours	550
16 hours	600
8 hours	650

Remove the panels from the furnace for as short a period of time as possible and inspect (except for the knife test) for compliance with 3.3.5 at the end of each heating period.

[1/] An apparatus of this type, powered by a 1/4 hp motor, operates at a rate of 1350 shakes per minute is manufactured by Red Devil Tools, Irvington, New Jersey (or equal).

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Remove two panels at the end of the 260°C, 320°C, and 500°C heating periods for use in the salt spray test (see 4.3.11). At the conclusion of the heating schedule remove the remaining two panels, allow to cool in air, cut with a knife blade, and observe for compliance with 3.3.5. Emissivity will be calculated in accordance with 4.3.16 for those two panels heated to 650°C, cooled to 260°C in the furnace, and held at 260°C for a period of four days.

4.3.10.2 Cold rolled steel. Spray the paint to a dry film thickness between $20.3X10^{-3}$ mm and $25.4X10^{-3}$ mm on two 76 mm by 152 mm steel panels that have been solvent cleaned as in 4.3.10.1. Apply the paint to the side of the panel that has been flat polished as described in note 1 to method 2011 of FED-STD-141. Air dry 30 minutes and then bake at 202°C ± 2°C for 1 hour. Cool to room temperature. Place the panels on a rack as in 4.3.10.1 and heat for 24 hours in an oven that has been previously raised to a temperature of 550°C ± 4°C. Remove from the oven, cool, and inspect for compliance with 3.3.5 performing the knife test as in 4.3.10.1.

4.3.11 Salt spray resistance.

4.3.11.1 Hot rolled steel. Expose the unscored panels removed from the 260°C, 320°C, and 500°C heating periods of 4.3.10.1 to the salt spray for 24 hours in accordance with method 6061 of FED-STD-141. Upon removal, wash the panels gently in warm running water until free from any visible salt deposits and examine immediately for compliance with 3.3.6.

4.3.11.2 Cold-rolled steel. Prepare three 101 mm by 305 mm steel panels as in 4.3.10.2 except do not heat at 550°C. Expose the unscored panels to the salt spray as in 4.3.11.1 for 96 hours. Upon removal, wash the panels gently in warm running water until free from any visible salt deposits and examine immediately for compliance with 3.3.6. Strip the paint from the panels by means of lacquer thinner and inspect for compliance with 3.3.6.

4.3.12 Water resistance. Prepare a film of paint as in 4.3.10.2 except do not heat at 550°C. Coat all exposed uncoated metal surfaces with wax and immerse the panel for 24 hours at 23°C ± 1°C in accordance with section 5D of ASTM D 1308-57. Upon removal, observe the panel for compliance with 3.3.7.

4.3.13 Hydrocarbon fluid resistance. Prepare a film of paint as in 4.3.10.2 except do not heat at 550°C. Immerse the panels for 4 hours in a hydrocabon fluid conforming to type III of TT-S-735 in accordance with method section 5D of ASTM D 1308-57. Upon removal, examine for compliance with 3.3.8.

4.3.14 Weather resistance. Prepare two 10.15 cm by 30.50 cm steel panels as in 4.3.10.2 except do not heat at 550°C. Place on outdoor exposure for 12 months at an angle of 45 degrees south in the latitude of Washington, D.C. in accordance with ASTM D 1014-66 and inspect for compliance with 3.3.9. Strip the paint from the panels by means of lacquer thinner, or if the thinner is not effective, by a commercial silicone enamel stripper and inspect for compliance with 3.3.9.

4.3.15 Toxicity. The manufacturer shall certify that the paint contains no benzene (benzol), methanol, chlorinated solvents or other ingredients which are deemed to be toxicologically hazardous under normal conditions of usage.

4.3.16 Emissivity. Normal emissivity shall be determined in accordnace with the following basic procedure; (refer to appendix A for detailed, in-depth theoretical derivation, equipment description, procedure, and a practical example):

(1) Emissivity is defined thusly:

$$\bar{\epsilon} = \frac{W(T_s') - W(T_b')}{W(T) - W(T_b')}$$

(With all temperatures expressed in degrees celsius)

Where:

$W(T_s')$ = Power radiated from a blackbody at apparent surface temperature, T_s'.

$W(T_b')$ = Power radiated by surrounding background at apparent temperature T_b'.

$W(T)$ = Power radiated by surface at actual temperature (T).

(2) Radiance, N(T), is a function of W at temperature (T) defined;

$N(T) = W(T)/TT$

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(3) The values $N(T)$ at (T) may be obtained from figure 2 and inserted in the equation of (1) above and $\bar{\epsilon}$ may be determined.

4.3.17 Flash point. Flash point determination is made in accordance with ASTM D 93-73. The flash point shall be in accordance with 3.3.12.

4.4 Inspection of preparation for delivery. Sample packages and packs and the inspection of packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Packaging, packing and marking. Paint shall be furnished in 1-quart or 1-gallon multiple friction closure type cans or in 5-gallon lug can or steel pails as specified (see 6.2). Packaging shall be Level A, B, or C as specified (see 6.2); packing Level A, B, or C as specified (see 6.2) and marking shall be in accordance with PPP-P-1892.

5.2 Special markings. In addition to the markings required by PPP-P-1892, interior containers shall be identified in accordance with FED-STD-313 and change(s) thereto. Each container shall also be marked or labelled with the following statement:

"CAUTION - USE ONLY WITH ADEQUATE VENTILLATION. AVOID CONTACT WITH SKIN AND PROLONGED BREATHING OF VAPOR"

5.3 Material Safety Data Sheets. A copy of the Material Safety Data Sheet, DD Form 1913, of FED-STD-313 shall be attached to each shipping container and palletized load for each shipping destination.

6. NOTES

6.1 Intended use. The paint covered by this specification is intended for use on superheated steam lines, boiler casings, boiler drums, superheater headers, and similar high temperature applications. It is also intended for painting military equipment such as personnel heaters, rocket launchers and other components, where operating temperatures preclude the use of conventional paints. It may also be used for application to engrave, stamp, or stencile lettering or numerals on metallic gun or weapon components for identification or functioning purposes when those components are exposed to temperatures up to 650°C.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of container required (see section 5).
(c) Level of packaging and packing required.

6.3 The paint should be purchased by volume, the unit being one United States liquid gallon of 231 cubic inches at 20°C.

6.4 Air pollution requirements. Air pollution characteristics of composition L paint shall be in conformance with air pollution Rule 66; Los Angeles County Air Pollution Control District.

6.5 Sub-Contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
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APPENDIX

EMISSIVITY MEASUREMENTS OF DIFFUSE, HIGH TEMPERATURE, OPAQUE SURFACES

10. SCOPE

10.1 Scope. This method describes a procedure for determining the average normal (90 degree) emissivity ($\bar{\varepsilon}$) of opaque surfaces in a spectral band ($\Delta\lambda$) by means of an infrared radiation detector and a surface thermometer. The following equation is used to define the average emissivity in the spectral band:

(1) $$\bar{\varepsilon} \equiv \frac{\int_{\Delta\lambda} J(\lambda,T)\,\varepsilon(\lambda)\,d\lambda}{\int_{\Delta\lambda} J(\lambda,T)\,d\lambda}$$

Terms in this equation are defined as:

$J(\lambda,T)$ Planck's blackbody radiation
$\varepsilon(\lambda)$ Spectral emissivity of sample surface

10.2 The method described is intended for measurements of diffuse and opaque surfaces where the surface temperature is much greater than the surrounding environment (background).

20. THEORY

20.1 The radiant power from a surface is the sum of the energies emitted and reflected from the surface. This is shown in equation (2) as:

(2) $$W(T_s') = W_\varepsilon + W_\rho$$

Where: $W(T_s')$ = Power radiated from a blackbody with surface temperature T_s' (W/cm^2)
W_ε = Power emitted by surface (W/cm^2)
W_ρ = Power reflected by surface (W/cm^2)

The relation:

(3) $$\int_{\Delta\lambda} J(\lambda,T_s')\,d\lambda = \int_{\Delta\lambda} J(\lambda,T)\,\varepsilon(\lambda)\,d\lambda$$

defines the apparent suface temperature (T_s') where T is the actual surface temperature.

20.2 For an opaque body

(4) $$\rho + \varepsilon = 1$$

Where ρ is the reflectivity and ε is the emissivity. The reflected power, W_ρ can be written as $\rho W(T_b')$ where $W(T_b')$, the power radiated from the background, is the power radiated by a blackbody at temperature T_b'. The apparent background temperature, T_b', is defined by equation (2) where the subscript (s) refers to the background (b), the temperature (T) is the actual background temperature, and $\varepsilon(\lambda)$ is the background emissivity. The power emitted from the surface, W_ε, equals $\bar{\varepsilon}\,W(T)$ where $W(T)$ is the power emitted from a blackbody at temperature T. The total power from the surface can thus be written as:

(5) $$W(T_s') = \bar{\varepsilon}\,W(T) + (1-\bar{\varepsilon})\,W(T_b')$$

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Solving the equation for $\bar{\varepsilon}$ gives:

$$\bar{\varepsilon} = \frac{W(T_s') - W(T_b')}{W(T) - W(T_b')} \qquad (5a)$$

30. EQUIPMENT

30.1 Two instruments are required for obtaining the required data. A radiation detector (such as the Barnes PRT5 precision radiation thermometer or equal) is needed to find the apparent surface (T_s') and background (T_b') temperatures and a contact thermometer is necessary for finding the actual surface temperature (T) of the sample.

30.2 A heat source is required for heating the sample should be capable of raising it to temperatures greater than 60°C above the ambient background temperature.

30.3 A schematic of the equipment is shown in figure 1.

40. PROCEDURE

40.1 To determine the normal emissivity in a spectral band, $\Delta\lambda$, it is necessary to find values of $W(T)$, $W(T_s')$, and $W(T_b')$ in that band. These are obtained by finding the apparent and actual surface temperatures of the sample, and the apparent temperature of the background.

40.2 A sample is placed on a heater and its temperature raised to greater than 60°C above the background temperature. After a stable temperature has been reached, determine the value of T with a contact thermometer. Remove the contact thermometer and determine T_s' and T_b' with a radiation thermometer. Repetition of this procedure for different values of T permits calculation of an average of the computed emissivities.

40.3 From temperatures obtained, the effective blackbody radiant power values $W(T)$, $W(T_s')$, and $W(T_b')$ can be determined by using a curve of effective blackbody power, $W_{eff}(T')$, versus temperature. This curve, supplied by the radiometer manufacturer or calculated, is found from the relation:

$$W_{eff}(T) = \int_{\Delta\lambda} J(\lambda, T) \frac{R(\lambda)}{R_{pk}} d\lambda$$ where

$R(\lambda)/R_{pk}$ is the normalized peak spectral response of the radiometer. The effective blackbody radiance curve for the PRT-5 precision radiation thermometer used is shown in figure 2. Blackbody radiance, $N(T)$, is related to the radiant power by the relation $W(T) = N(T)\pi$.

40.4 Sample calculation. The following is a sample calculation for the temperatures: T = 100°C, T_s' = 75°C, and T_b' = 25°C. The respectivie values of radiance are found from figure 2 to be; N(100) = 11.55, N(75) = 8.85 and N(25) = 4.55

Therefore, $$\bar{\varepsilon} = \frac{8.85 - 4.55}{11.55 - 4.55} = 0.61$$

50. PROCEDURE VERIFICATION

50.1 The emissivities of samples determined by the designated procedure were compared to data taken with a Gier-Dunkle or equal heated cavity reflectrometer. A summary of the results, showing averaged values, is given in table III; good agreement between the two methods was obtained.

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TABLE III. Emissivity measurement summary.

Sample No.	Emissivity, $\Delta\lambda$ = 8-14 μm	
	Described Procedure	Gier Dunkle Heated Gravity
06-527	0.64	0.61
06-535	0.91	0.91
Black Velvet	0.93	0.93-0.95
1	0.55	0.46
2	0.53	0.46
08-683	0.49 1/	0.46

1/ Data for $T_s > 60°C$.

60. ERROR ANALYSIS

60.1 A complete error analysis is beyond the scope of this task. However, examples will be given which show how instrument inaccuracies affect the calculated emissivity. Data in 4.4, used as a reference, will be changed and the corresponding change in emissivity calculated. Results are shown in table IV.

TABLE IV. Error analysis.

T	$N(T)$	T_s'	$N(T_s')$	T_b'	$N(T_b)$	ε	$\Delta\varepsilon$
100	11.55	75	8.55	25	4.55	0.61	REFERENCE
98	11.32	75	8.55	25	4.55	0.635	+.025
102	11.78	75	8.55	25	4.55	0.595	-0.02
100	11.55	73	8.7	25	4.55	0.593	-0.007
100	11.55	77	9.1	25	4.55	0.593	+0.04
100	11.55	75	8.55	23	4.44	0.62	+0.01
100	11.55	75	8.55	27	4.71	0.605	-.005

60.2 For the example shown in section 60.1 (which is a typical case) a change in T of $\pm$ 2°C causes a corresponding change in calculated emissivity of 0.045. A change in T_s' of $\pm$ 2°C causes a change of 0.047 and T_b' $\pm$ 2°C results in a 0.015 change in the calculated emissivity.

60.3 The change in emissivity that occurs when both T and T_s' are varied can also be found from values in table I. For T = 102°C and T_s' = 73°C the resulting change in emissivity is found to be 0.045.

60.4 Data in sections 60.1 and 60.2 indicate that equipment accuracies of $\pm$ 1°C would be more than sufficient for insuring that the calculated emissivity is within the range required, which is $\pm$ .05 emissivity units.

70. SUMMARY AND CONCLUSION

70.1 A simple, direct method is presented for measuring the emissivity of heated surfaces. The surface temperature should be above 60°C during the measurement. Measurements are made with both radiometer and contact thermometer. Emissivity measurements with an error less than 0.05 emissivity units are possible.

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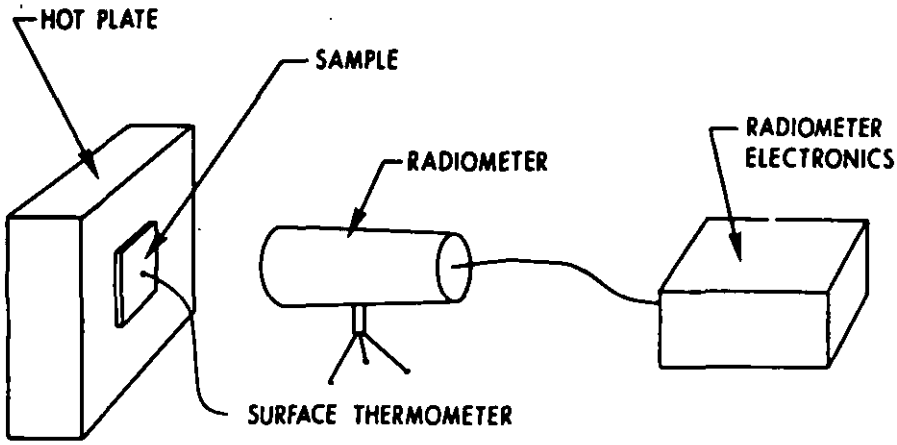


FIGURE 1. Equipment configuration.

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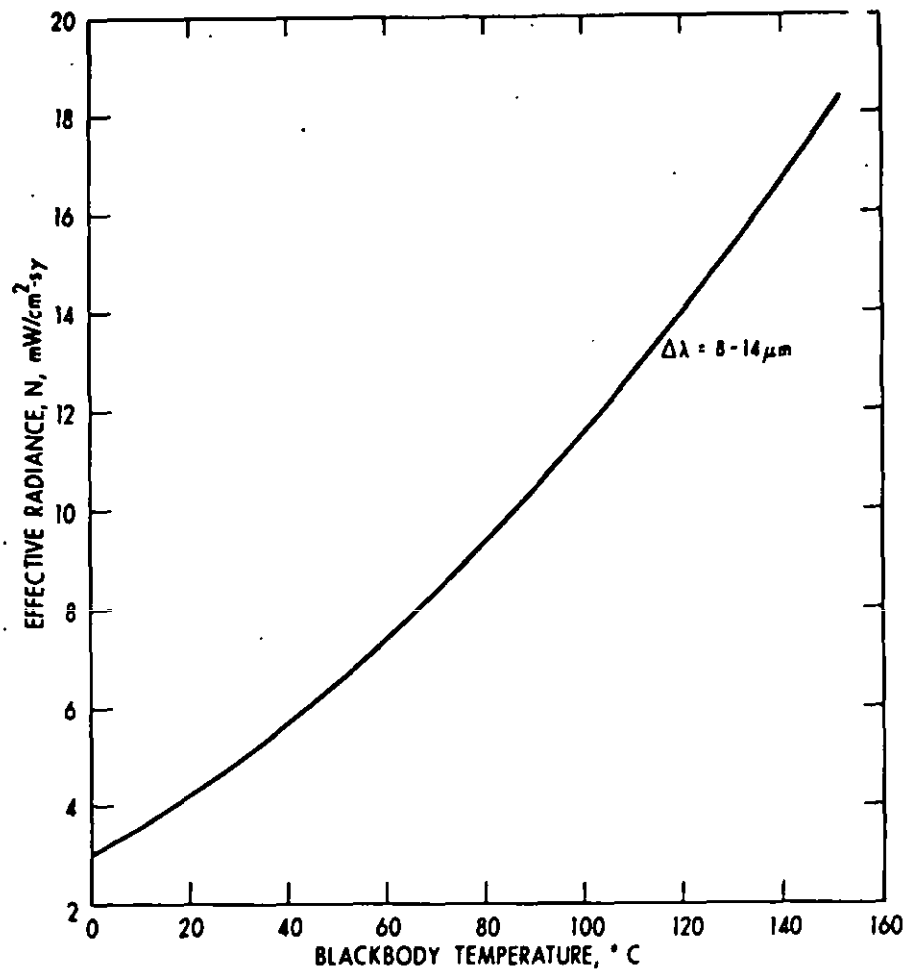


FIGURE 2. Effective blackbody radiance for Barnes PRT-5 S/N 227 (or equal).

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